Question 1

The real estate profession commonly uses discounted cash flow (DCF) methods to value real estate assets and investment opportunities.

a- How does a DCF method combine risks and opportunities? And what are the most important risk and opportunities to consider when *directly* investing in real estate? (5 points)

These issues were discussed at length during the first part of Lecture 1.

b- What other methods are used to value real estate assets? Consider both the approaches discussed during the regular lectures as well as the insights provided by the external speakers. (5 points)

Methods such as comparables or replacement costs were discussed during Lectures 1 and 2, as well as during Lecture 6 by the Tristan Capital external speakers.

c- How can a pro-forma analysis help investors assess the sources of revenues and costs that should be considered when putting together a DCF model? (5 points)

The second half of Lecture 2 provided a discussion of these topics.

d- Rent roll and operating costs. Discuss the main considerations that need to be kept in mind when forecasting these quantities in your pro-forma calculations. Use the materials discussed during the main lectures as well as the insights you gained from working on the two case-study assignments for the course. (15 points)

These points were covered during Lecture 2. I expect students to discuss various ways of calculating rents per square foot in ways that adjust for void periods, concessions, overage charges, etc. I also expect students to discuss square footage measurement and charges for communal areas. Several points were also made in terms of type of contracts that can be written to pass down some of the operating costs to the tenants. Lastly, I hope the students will use both the Croydon and Leeds case studies to enrich their answers.

e- What is the most common approach to estimate the resale value of a building at the end of the holding period? How can you ensure that the building is 'valuable' and sold at premium relative to its purchase price? (10 points)

Lecture 2 and 3 discussed these points (exit yields or cap rates).

f- Discuss some commonly used risk-assessment methods for direct real estate investment. What aspects of the DCF calculations are normally 'tested'/'stressed', and with what aims? (10 points)

These issues were discussed during Lecture 3, as well as in the context of the case studies where students were asked and shown how to perform sensitivity analysis that stress the most important elements of their cash flow projects.

g- Consider structuring a portfolio of direct real estate investments. What are the most important data limitations when assessing its possible performance and risk-return profile? What methods have been used to overcome such limitations? (20 points)

h- What are the most important considerations to keep in mind when choosing assets to structure a portfolio of direct real estate investments? Consider at least the following issues: liquidity, asset characteristics, number of properties. (10 points)

These issues raised at points g- and h- were the topics of the external lecture delivered in Week 9 by Dr. Niko Szumilo. They are well summarised in the set of slides he distributed and was made available on Moodle. He also had an opportunity to deliver a synchronous lecture, on top of the narrated power point, both available on Moodle.

i- Environmental sustainability issues are becoming more important when considering the value, costs and risks of real estate assets. Discuss how 'green' buildings can add value for investors and mitigate some of these costs and risks. Consider both the point of view of a direct investor (i.e., someone planning to own an environmentally sustainable building) as well as the perspectives of a potential lender (i.e., someone lending against a 'green' building). (20 points)

These issues were discussed in lecture 9 by Dr. Liao (narrated power point) and in the panel discussion that took place during the synchronous part of that lecture (including Dr. Liao, Prof. Silva and two real estate finance practitioners).

Question 2

You are planning to borrow £20,000,000 to invest in a building worth £40,000,000. Your expected holding period is five years and you expect revenues from the building to accrue to you quarterly. Consistently, the loan structures you are discussing with the bank have quarterly payments and quarterly compounding. The annual interest rate the bank is planning to charge is 5%.

a- Consider an interest-only (IO) loan. What is the value of the quarterly payments you owe to the bank? And what is the size of the balloon payment at the end of Year 5? (5 points)

The loan of 20,000,000 would be charged a quarterly interest of 5%/4=1.25%. The quarterly payment would therefore be 250,000 while the balloon payment at the end would be for the full amount of the loan – at 20,000,000. Such calculations were discussed at several points in lectures 3, 4, and 5.

b- Consider instead a constant payment mortgage (CPM). The loan charges an annual interest of 5%, and is structured over 30 years with quarterly payments and compounding. What is the value of the quarterly instalments you need to pay to the bank? What is the outstanding balance on this loan at the end of Year 5? (5 points)

Constant payment mortgage calculations were discussed during lecture 4 and 5. In this case, the students need to set compounding to 4 (quarterly) and enter data as follows: N=30 (since there is compounding at 4, this is transformed into 120), I/Y=5% (the calculator automatically translates into the quarterly equivalent); PV=-20,000,000. The quarterly PMT of this loan is 322,669.91 with outstanding balance at the end of year 5 (period 20) = 18,360,350.29. This can be found with the AMORT worksheet setting the second period to 20 (5 years with quarterly compounding).

c- The bank has also informed you that both loans carry an origination charge of £200,000. What is the effective cost of borrowing for the IO loan if you reimburse it at the end of Year 5? And what is the effective cost of borrowing for the CPM loan if you repay at the end of Year 5? Compare your findings and discuss what you notice. (10 points)

The origination charge of 200,000 means that the loan payments are calculated on the full loan but the borrower actually receives 19,800,000. For the IO loan, the effective cost of borrowing can be calculated using the cash flow worksheet on the calculator as follows. CF0=-19,800,000; CF1=250,000 for 19 periods (all quarters except the last); CF2=20,250,000 (loan reimbursement plus regular interest payment) for one period. This gives a quarterly IRR that when multiplied by four gives an annual IRR of 5.2286%. In the case of the CPM, similar calculations need to be carried out. But the CF1 would be 322,669.91 while CF2 would be 18,360,350.29 (the outstanding balance) plus 322,669.91 (the regular payment). This gives an annual effective cost of borrowing of 5.2370%. This is larger because the amortizing feature makes the initial CPM payment larger than those of the IO, and these weigh more from the point of view of discounted present values that later payments.

d- What is the rationale for banks to impose origination charges? What do banks 'like' in a CPM loan compared to an IO loan? Why might borrowers instead prefer an IO loan? (5 points)

These issues were extensively discussed in Lecture 5.

e- Consider now an alternative adjustable rate mortgage (ARM) with quarterly payments and compounding. The mortgage charges a 4% annual interest rate for the first two years – before resetting at the beginning of Year 3 and again at the beginning of Year 5. You expect the annual interest rate you will be charged at the beginning of Year 3 to be 6%, before a further climb in Year 5 to 7%. What are the payments for this loan in the three periods when different rates prevail? And what is the final contribution to fully reimburse the loan at the end of Year 5? (20 points)

ARM loans can be considered as sequences of CPM loans. The first loan would be for the full 30 years, charging 4% on the full loan of 20,000,000 with quarterly compounding. This would give an payment of 286,941.8968. After two years (8 periods), using the AMORT worksheet students can find the outstanding loan is 19,279,628.09. This is the new principal for the loan charging 6% from the beginning of year 3 for the remaining 28 years. This gives a payment of 356,464.0452 and after two years (8 quarters) the outstanding balance of this loan would be 18,712,354.18. This is now the new principal for the loan charging 7% for 26 years. The payment is 391,985.6568. After 1 year (4 quarters), we reach the end of the holding period and the loan is reimbursed. The outstanding balance is 18,447,422.42.

f- Calculate the effective cost of borrowing of this loan considering that it still carries an origination charge of £200,000. How does this compare to the effective costs for the loans discussed at points a-c? What do you notice? (10 points)

To calculate the effective cost of borrowing of this loan, the students need to use the cash flow worksheet with the following inputs. CF0=-19,800,000. CF1= 286,941.8969 for 8 quarters; CF2=356,464.0452 for another 8 quarters; CF3=391,985.6568 for 3 quarters; and finally CF4 =391,985.6568 + 18,447,442.42 (the outstanding loan) in the final quarter. This gives an annual IRR (or effective cost of borrowing) of 5.5256%. These calculations were covered in Lecture 5 (using annual compounding and a simpler example – but past exams contain plenty of such calculations for the students to practice).

g- Consider now that the building will generate a quarterly NOI of £275,000 for the first two years, climbing to £500,000 from the beginning of Year 3 to the end of Year 5. You also expect to sell the building at end of your holding period for £45,500,000. What is the annual IRR on this investment without leverage? (10 points)

Such calculations were covered during the first two weeks of the course. In this case, the students have to use the cash flow worksheet with the following inputs. CF0=-40,000,000. CF1=275,000 for 8 quarters; CF2=500,000 for 11 quarters; CF3=500,000+45,500,00 for one quarter. The implied annual IRR is 6.3708%.

h- Now calculate the three leveraged annual IRRs using the three loan structures discussed above (IO, CPM and ARM). What do you notice? Why do you think some loans give more/less favourable results? In answering this question, consider the times when potential losses/revenues accrue to the investor. (25 points)

Starting with the IO loan, the data for the cash flow model would be as follows. CF0=-20,200,000 – noticing that 200,000 have to be put in by the equity holder as the bank is charging a 200,000 origination fee. Next, there would be CF1=275,000-250,000 for 8 quarters; then CF2=500,000-250,000 for 11 quarters and finally CF3=500,000-250,000+45,500,000 – 20,000,000 (notice that even if the bank only gives 19,800,000 the balloon repayment for loan at the end is for the full amount). This gives an annual IRR of 7.3602%.

For the CPM mortgage the data are as follows. CF0=-20,200,000; CF1=275,000 – 322,669.91 for 8 quarters. Note that these negative initial cash flows will weigh down the performance of the investment because losses are front loaded. CF2=500,000-322,669.91 for 11 quarters. And finally CF3=500,000-322,669.91+45,500,000 – 18,360,350.29 (the outstanding loan, remember the CPM is amortizing) for 1 quarter. This gives an IRR of 7.2906% lower than with the IO because of the early losses.

Finally, for the ARM the calculations are as follows. CF0=-20,200,000. CF1=275,000-286,941.8962 for eight periods. CF2=500,000-356,464.0452 for another 8 periods. CF3=500,000 – 391,985.6568 for 3 quarters and finally CF4=500,000-391985.6568 + 45,500,000 – 18,447,422.42 for the last quarter. This gives an IRR of 7.0732%. This is the lowest because even if the ARM charges little at the beginning the later spikes in interest charges give a higher overall effective cost of borrowing than the other two loans.

i- What would have been your estimate of the leveraged returns if you had used a Weighted Average Cost of Capital (WACC) approximation? How do your answers differ from those found at point h-? And what does this approximation 'miss' that instead you are able to capture with a fully-fledged DCF approach? (10 points)

The students could use a WACC based on the nominal cost of borrowing of the three loans (5% for the IO and CPM loan and 4% at the beginning for the ARM) and compare this to the non-leveraged IRR of 6.3708% with a D/E=1 (20,000,000/20,000,000). This would suggest that the IO and CPM give the same return on equity and the ARM is the best option. But this would be wrong as the right solution takes into account the effective cost of borrowing. So the right numbers would be as follows:

For the IO: 6.3708% + (6.3708% - 5.2286%) * 19,800,000/20,200,000 (=0.9802) → 7.49%

For the CPM: 6.3708% + (6.3708% - 5.2370%) * 19,800,000/20,200,000 (=0.9802) → 7.4822%

For the CPM: 6.3708% + (6.3708% - 5.5256%) * 19,800,000/20,200,000 (=0.9802) → 7.1987%

The "ranking" in terms of what loan performs better using the WACC are correct, but the figures are not. This is because the ratio D/E changes over time as a function of equity changing (changing value of the asset) and amortization contributions (for CPM and ARM).